2024 International Conference on Wireless Communications Signal Processing and Networking (WiSPNET) | 979-8-3503-5084-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/WISPNET61464.2024.10532954

Enhancing Forecasting Precision in Indian Industrial Workers' Inflation: A Comparison of Advanced Machine Learning Models

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***Abstract*—Inflation is a key economic indicator of a country's growth in the contemporary world. This research paper focuses on time series analysis and forecasting of inflation data for Indian industrial workers by employing various machine learning models. This study utilizes the Consumer Price Index (CPI) to measure inflation. It explores and compares the application of Long Short-Term Memory (LSTM), bidirectional LSTM (BiLSTM), XGBoost, and Random Forest models for predicting future inflation values. The dataset used for this spans from January 1993 to November 2023, sourced from the Labour Bureau of the Government of India. The models are evaluated based on mean squared error (MSE) and their ability to forecast the inflation value for November 2023. The proposed BiLSTM model outperforms the others in terms of MSE and accuracy in forecast values. Although computationally intensive, BiLSTM demonstrates enhanced context understanding and improved learning for long-term dependencies in time series data with a MSE value of 0.9410. The study emphasizes the effectiveness of various forecasting models and the use of data-driven approaches to make decisions based on empirical evidence and inflation trends. It also highlights the potential for incorporating AI into economic analyses for continuous monitoring, effective policy-making, and future advancements.**

***Index Terms*—Consumer Price Index, Inflation, Time Series Analysis, Forecasting**

I. Introduction

Inflation is the price rise leading to a decline in consumer purchases over time. When there is an increase in the supply of money, the demand for goods also soars up. This exceeds the production capacity of the economy. The increased demand, in turn, causes the prices to go high. This is the leading cause of inflation in most sectors and results in a cost-push effect and built-in inflation where there is a demand for higher wages for workers to meet the increased demand for goods. Different indexes are used to calculate inflation, like the Consumer Price Index(CPI), Producer Price Index(PPI) and Wholesale Price Index(WPI). [1]

Consumer Price Index more commonly known as CPI, is a measure used to calculate a country's inflation in its economy. It measures the monthly changes in the price paid by customers. The CPI is computed as the cost of the market basket in the necessary year divided by the cost of the market basket in the base year. India's CPI data has been compiled and maintained by the Labour Bureau since 1944. There are four types of CPI in India. The CPI for industrial workers measures the change in prices of fixed goods and services used by industrial workers over a while. It covers employees from industries like workers from ports and docks, mines, railways, plantations, electricity generation, factories and electricity generation. The CPI for agricultural and rural labourers depends on the monthly price data collected from villages. CPI data for urban non-manual employees is derived from employees who obtain more than 50% of their income from employment on non-manual work in an urban sector that is not based on agriculture. [2] [3]

Recent technological advancements in AI have aided the Finance Sector in many ways. [4] Researchers continue to exploit various machine learning and deep learning methods for various economic and financial tasks. Time series analysis is the process of tracking a succession of data points gathered over a given period. It is used to analyse how data changes over time and make important conclusions. It aids in identifying the underlying trends, causes, and patterns in the data across time or for a certain component causing change. Forecasting is the prediction of future values based on given facts. Machine learning techniques may be used to analyse time series data by making predictions and forecasts. Popular models used for this purpose include Autoregressive (AR), Moving-Average (MA), Autoregressive Integrated Moving Average (ARIMA) and Long Short-Term Memory (LSTM). This study focuses on LSTM, Bidirectional LSTM (BiLSTM), XGBoost, and Random Forest for time series analysis of Indian industrial workers' point-to-point inflation data, forecasting future values, and selecting the optimal model based on mean squared error, training, and validation loss. [5] [6]

II. Literature Review

Governments and businesses primarily use CPI to monitor the price change in an economy. Analyzing the trends in the CPI data of an economy helps identify inflation rates and cater to people's needs accordingly. Since it is measured monthly, data monitoring can get complex and tedious. Hence, time series analysis can aid in predicting and forecasting the data. Kharimah et al. did time series modelling and forecasting on Bandar Lampung's CPI data. The study utilises time series graphs, Autocorrelations function (ACF), and unit root testing [7]. The model is derived from ACF and the Partial Autocorrelations function (PACF). ACF measures the similarity between time series data and a lagged version. It correlates a data point's current value and past values. The authors use metrics like mean squared error(MSE), bayesian information criterion(BIC), and Akaike Information Criteria(AIC). The best model was identified as ARIMA(1,1,0). It is the differenced first-order autoregressive model where the first difference of the dependent variable is lagged by one period.

Time series analysis is used to analyze economic indicators of various countries. Gjika et al. explored the relationship between CPI and economic indicators like exchange rate and the number of people traveling abroad. They provided an understanding of the economic and social development of the country. The paper used multiple regression models and time series forecasting models. The SARIMA model with ETS produced good findings for each subcomponent's connection to the CPI, and the multiple regression model is employed to forecast CPI. They conclude that the stability of the European currency(euros) and the number of people traveling abroad result in Albania's CPI value growth. [8] Similarly, a study on the CPI changes in the United States economy indicated that the Multivariate Adaptive Regression Splines (MARS) algorithm is the best model for accurately forecasting the CPI values. The study provides a detailed analysis of the CPI trends to policy makers and investors for better economic decisions. [9]

A paper discussing the trends in Ukraine's CPI values from January 2010 to September 2020 uses the box-Jenkins model, the additive ARIMA*ARIMAS model, and an exponential smoothing model with the Holt-Winters seasonality estimate. The inflation forecast was based on the Holt-Winters model, which models the three aspects of time series: average, trend, and seasonality. They also considered the COVID-19 pandemic and concluded that it did not affect the CPI rate. There was a constant 5-10% growth in the inflation level every year in the Ukrainian economy. [10] Kim, T. H., et al. compare the relative temporal trends of good and unhealthy eating trends in South Korea from 1995 to 2015. The study uses time series analysis to investigate and forecast pricing changes in several food categories such as cereals, vegetables, meats, sweets, spices, fast food, and non-alcoholic drinks. The authors discovered that the price of processed meat climbed by 1.2%, beef by 2.4%, milk by 1.4%, and the annual increase in fast food products was smaller than the CPI. [11] Doho et al. discuss the effect of sectoral indices on inflation in the west African regions. The study concludes that utlities and agricultural sectors have a bigger impact on inflation using an associated kernel method. It is also discovered through this study that relationship between stock market indices and inflation is non linear. [12]

Encoder decoder models like LSTM and BiLSTM are also used for time series forecasting. Siami-Namini et al. compare the results of three models, ARIMA, LSTM, and BiLSTM, for financial time series modeling and forecasting. The results are evaluated based on Root mean squared error(RMSE) and percentage reduction metrics. The paper states that conventional models like ARIMA perform better on short-term forecasts but degrade for long-term predictions. The average RMSE value of BiLSTM is 20.17 compared to LSTM, which is 39.09, and ARIMA, which is 302.96. The authors evaluate the learning procedures by studying, how LSTM and BiLSTM models learn. The study found that BiLSTM outperformed LSTM with a 37.78% decrease in the error rate. [13] [14]

Other deep learning models like XGBoost have been used for time series modeling issues like network traffic in telecom networks and stock price forecasting. [15] Y. Wang et al. used a hybrid model with ARMIA and XGBoost for stock price forecasting; after an experimental comparison of 10 stock datasets, the hybrid model performed better than the prediction of four models of ARIMA, XGBoost, GSXGGB, and DWT-ARI-MA-XGBoost. [16]

III. Methodology

A. Exploratory Data Analysis

The dataset is obtained from the Government of India's Labour Bureau website. The data contains point-to-point inflation data for industrial workers' Consumer Price Index (CPI) values by calculating the relative change in prices of goods and services used by industrial workers. With 88 centers across the country, this data is collected through surveys on the price fluctuations of 31 commodities on a monthly basis. This data is primarily used for regulating the wages and allowance of workers in the country. It is also used for measuring inflation and policy formulations.

The Labour Bureau conducts surveys of the prices of 31 commodities at 88 urban centers across the country. The data has been recorded from January 1993 to November 2023. The dataset has 372 entries with the base year, current year, month, and point-to-point inflation data. The data is preprocessed into a pandas data frame and is split into 80% for training and 20% for testing for all the models. The plot of inflation data versus the date is shown in Figure 1.

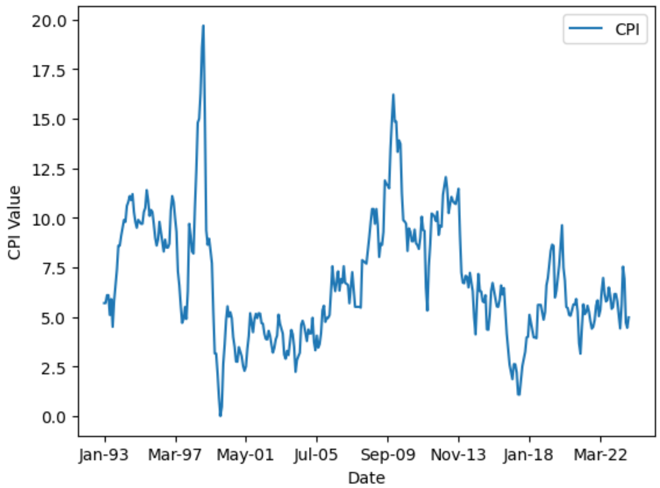

Fig. 1: Industrial workers' inflation data over the years

B. LSTM

LSTM is a Recurrent Neural Network (RNN) that learns long-term dependencies and avoids them. RNNs employ a single tanh layer for its repeating module, but LSTMs utilise four neural network layers. [17] They contain a cell state that is controlled by gates, and the LSTM cannot add or delete data from it. They have three gates: an input gate that decides whether information should be stored in the cell state and controls the flow of new information into the cell state. The next gate selects the information to be deleted, and the final output gate controls the information passed to the next character in the sequence. It sets the prediction parameters for the cell state. [18]

LAYER	OUTPUT SHAPE	PARAMETERS
Input Layer	(None, 1, 1)	0
LSTM Layer	(None, 1, 256)	264192
Batch Normalisation	(None, 1, 256)	1024
Dropout Layer	(None, 1, 256)	0
LSTM layer	(None, 1, 128)	197120
Batch Normalisation	(None, 1, 128)	512
Dropout	(None, 1, 128)	0
LSTM Layer	(None, 64)	49408
Dense Layer	(None, 1)	65

TABLE I: LSTM Model Summary

The proposed model contains an input layer and 3 LSTM layers. The first and second layers have 256 and 128 units, each with return sequences set to true, batch normalization, and dropout layers. This helps in normalizing by scaling and centering the input layers. The dropout layer prevents overfitting by randomizing the input units during training. The third layer has 64 units, and a dense layer with 1 unit and a linear activation function is attached to the output layer as seen in Table I. The optimizer used is Adam, with a learning rate of 0.001 and loss function used in mean squared error. Finally, the model provides future predictions based on previous predictions.

C. BiLSTM

A Bidirectional LSTM is an RNN layer that learns the long-term dependencies like a normal LSTM but between the step time of a time series or sequential data. [19] They aid in learning the complete time series at every step. It contains two unidirectional LSTM layers for processing input in two directions, i.e., forward and backward. The input sequence is given as it is to one of the layers while it is reversed for the other layer. [20] This way, every input component contains details about the past and current states. Each returns a probability vector for output, and a combination of individual output vectors gives the final output. Figure 2 represents the working of a BiLSTM proposed by Heshan et al. [21]

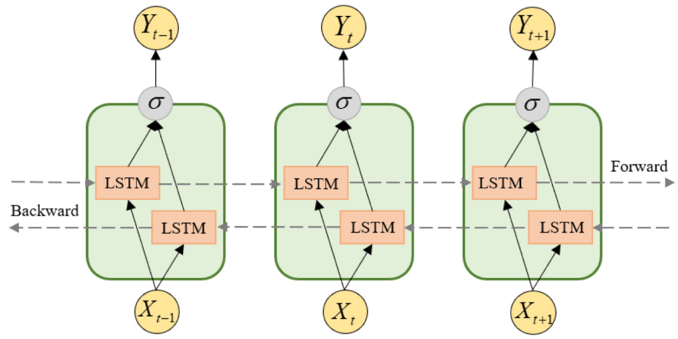

Fig. 2: The architecture of BiLSTM

The proposed model, similar to the LSTM model, has three layers. It contains an input layer, the first BiLSTM layer with 256 units, the second BiLSTM layer with 128 units, and the third BiLSTM layer with 64 units with batch normalization after the first BiLSTM layer and dropout layers as seen in Table II. It uses a linear activation function and Adam optimizer to minimize the loss during training. The model's performance is evaluated through mean squared error. It provides future predictions based on previously predicted values.

LAYER	OUTPUT SHAPE	PARAMETERS
Input Layer	(None, 1, 1)	0
Bi-Directional Layer	(None, 1, 512)	528384
Batch Normalisation	(None, 1, 512)	2048
Dropout Layer	(None, 1, 512)	0
Bi-Directional layer	(None, 1, 256)	656384
Dropout	(None, 1, 256)	0
Bi-Directional Layer	(None, 128)	164352
Dense Layer	(None, 1)	129

TABLE II: Bi-directional LSTM Model Summary

D. XGBoost

eXtreme Gradient Boosting is a machine-learning library designed for supervised tasks. It is a distributed gradient-boosted decision tree designed primarily for ranking, classification, and regression problems. It is based on a gradient-boosting framework where various decision trees are combined to create the predictive model. [22] XGBoost contains various advantages that make it a good model for tasks like time series analysis, considered a supervised learning task. XGBoost builds its trees depth-wise to capture all the essential features and also provides a feature importance score to understand the features that contribute the most to the model's predictions. It is also very efficient by supporting parallelization and scalability across multiple nodes of the training process.

E. Random Forest

Like XGBoost, Random Forest is also constructed of decision trees during the training process and produces the mean prediction of each tree as the output for regression or classification-based problems. [23] The random forest algorithm is based on the bagging method, which utilizes bagging and the randomness of features to create decorrelated decision trees. During the splitting process of training, the features are taken and considered as random subsets. The randomness is done to reduce overfitting, make the model more robust, and decorate the decision trees.It comprises three hyperparameters that must be configured before training. They are the node size, the number of decision trees employed, and the number of sample characteristics. For regression-based problems, the final prediction is calculated as the average of the predictions made by each tree. In contrast, classification problems produce results based on the majority class projected by the trees. [24]

IV. Result Analysis And Inference

A standard metric, mean squared error (MSE) is utilised to evaluate the models' performance. All of the models utilised generate the mean square error and the forecasted value as outputs. Mean squared error is a metric used to assess the performance of regression models. [25] It calculates the average squared difference between actual and predicted values. The formula of MSE is given in equation (1) where n is the number of data points, D_i is the actual value, and $\hat{D}_i$ is the predicted value for the particular data point.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(D_i - \hat{D}_i\right)^2 \tag{1}$$

MSE takes the squared differences between the predicted and actual values to penalize significant errors compared to minor ones. It is a positive value, and an MSE of 0 indicates no error, which is impossible. The lower the MSE value the better the performance of the model. [26] Apart from MSE, the forecasted value is also compared against the original inflation value of November 2023, which is 4.98.

The LSTM model was trained for 132 seconds, resulting in a loss value of 1.21 at training and a loss value of 0.714 at validation. The MSE value of the model is 0.992, as seen in Table III. This model was trained on data from January 1993 to October 2023. It forecasted the value inflation data of November 2023 as 5.0454, which is 0.06 more than the actual value. The model performed the second best for its MSE value and trained the fastest regarding training time. However, the LSTM model needed help capturing future data point patterns. To overcome this, a BiLSTM model is used since it can leverage information from both the past and future time steps.

The BiLSTM model trained for 336s with a training loss value of 1.14 and a validation loss value of 0.693. The MSE value of the model is 0.9410, and the forecast value it predicted is 5.008, which is 0.028 more than the actual inflation data of 4.98. The original versus predicted results of the model are shown in Figure 3. It can be seen that a very slight difference occurs in the graph indicating that the BiLSTM model performs best regarding MSE and accuracy in the forecasted values. Although it is computationally complex and takes more time to train, the model has an enhanced context understanding than the other models used and has improved learning for long-term dependencies in the time series data. Hence, it is the best model among the other trained models.

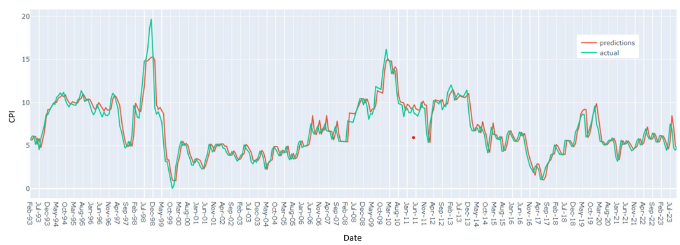

Fig. 3: Original vs Predicted results of the BiLSTM model

The XGBoost model has an MSE of 1.21, and the forecast value for November 2023 is 5.44, which is 0.46 more than the actual value. The model used lag features by shifting the CPI values in time to create a lag value representing the value from the previous time step. It aimed to capture the historical trends and patterns in the data and minimize the MSE between actual and predicted CPI values during training. The model was tested on 20% of the dataset. It shows some fluctuations in the actual and predicted value for the test data when there is a drastic increase or decrease in the CPI value, as shown in Figure 4. Towards the end, in the validation phase, the model performs poor compared to the training phase with significantly high difference in the acutal and predicted values. The Random Forest model has an MSE of 1.021 and 5.202. Although both the models are suitable for time series predictions, more complex models like LSTM and BiLSTM performed better since they captured the temporal patterns and sequential dependencies better than XGBoost and Random Forest.

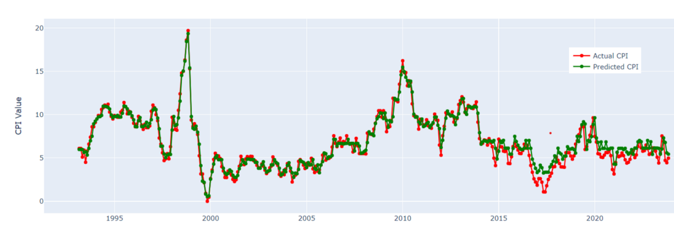

Fig. 4: Original vs Predicted results of the XGBoost model

Model	MSE value	Actual value	Forecast value
Long Short Term Memory(LSTM)	0.992	4.98	5.0454
Bidirectional Long Short Term Memory(BiLSTM)	**0.9410**	4,98	**5.008**
XGBoost	1.219	4.98	5.44
Random Forest	1.021	4.98	5.202

TABLE III: Evaluation Summary

V. Conclusion and Future Scopes

This paper delves into the time series analysis and forecasting of inflation data for Indian industrial workers. It has been observed that BiLSTM performs well compared to the other models used based on the Mean squared error metric and the forecasted value obtained as a result. To reduce complexity, the model parameters were chosen after analysing the parameter sensitivity by modifying the number of hidden layers, input units and optimizing the learning rate parameter. The other models can perform better and produce satisfactory results with more fine-tuning and hyperparameter tuning. Several factors like changes in consumption patterns of consumers, exchange rates, government policies and seasonal variations can affect the CPI values. The proposed models do not consider these factors and hence the future scopes for this research work include the usage of state-of-the-art models like Large language models(LLM) for prediction, reducing the computational complexity by carefully optimizing the model architecture, utilizing a combination of different models to create a hybrid model to give better results along with continued research and experimentation to create more accurate and robust models for predicting inflation data.

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